Q1: 16 March (Shift 2) - Numerical

A $5.0 \mathrm{~m} \mathrm{~moldm}^{-3}$ aqueous solution of KCl has a conductance of $0.55 \mathrm{mS}$ when measured in a cell constant $1.3 \mathrm{~cm}^{-1}$. The molar conductivity of this solution is ___ $\mathrm{mSm}^{2} \mathrm{~mol}^{-1}$.
(Round off to the Nearest Integer)

Q2: 17 March (Shift 2) - Numerical

A KC solution of conductivity $0.14 \mathrm{~S} \mathrm{~m}^{-1}$ shows a resistance of 4.19Ω in a conductivity cell. If the same cell is filled with an HCl

solution, the resistance drops to 1.03Ω. The conductivity of the HC1 solution is ____ $\times 10^{-2} \mathrm{~S} \mathrm{~m}^{-1}$. (Round off to the Nearest Integer).

Q3: 18 March (Shift 1) - Numerical

For the reaction

$2 \mathrm{Fe}^{3+}(\mathrm{aq})+2 \mathrm{I}^{-}(\mathrm{aq}) \rightarrow 2 \mathrm{Fe}^{2+}(\mathrm{aq})+\mathrm{I}_{2}(\mathrm{~s})$

the magnitude of the standard molar free energy

change, $\Delta_{\mathrm{r}} \mathrm{G}_{\mathrm{m}}^{\circ}=-$_____kJ (Round off to the Nearest Integer).

Q4: 18 March (Shift 2) - Numerical

The molar conductivities at infinite dilution of

barium chloride, sulphuric arid and hydrochloric acid are 280,860 and $426 \mathrm{Scm}^{2} \mathrm{~mol}^{-1}$ respectively. The molar conductivity at infinite dilution of barium sulphate is______

$\mathrm{Scm}^{2} \mathrm{~mol}^{-1}$(Round off to the Nearest Integer).

Answer Key

Q1 (14.3) **Q2** (57) **Q3** (45) **Q4** (288)

Q1 (14.3)

Given conc ^{n} of KCl $= \frac{\mathrm{m \cdot mol}}{\mathrm{L}}$

$\therefore$ Conductance $(\mathrm{G}) = 0.55\mathrm{mS}$

Cell constant $\left(\frac{\ell}{\mathrm{A}}\right) = 1.3\ \mathrm{cm}^{-1}$

To Calculate : Molar conductivity (λ_{m}) of sol.

$\rightarrow$

$\rightarrow$ Molarity $= 5 \times 10^{-3} \frac{\mathrm{mol}}{\mathrm{L}}$

$\rightarrow$ Conductivity $= \mathrm{G} \times \left(\frac{\ell}{\mathrm{A}}\right) = 0.55\mathrm{mS} \times \frac{1.3}{\frac{1}{100}}\ \mathrm{m}^{-1}$

$= 55 \times 1.3 \quad \mathrm{mSm}^{-1}$

$\mathrm{eq}^{\mathrm{n}}(1)\lambda_{\mathrm{m}} = \frac{1}{1000} \times \frac{55 \times 1.3}{\left(\frac{5}{1000}\right)} \frac{\mathrm{mSm}^{2}}{\mathrm{mol}}$

$\Rightarrow \lambda_{\mathrm{m}} = 14.3 \frac{\mathrm{mSm}^{2}}{\mathrm{mol}}$

Q2 (57)

$\kappa = \frac{1}{R} \cdot G^{*}$

For same conductivity cell, G^{*} is constant and hence $\kappa \cdot R. =$ constant. $\therefore 0.14 \times 4.19 = \kappa \times 1.03$

or, $\kappa = \frac{0.14 \times 4.19}{1.03}$

$= 0.5695\mathrm{Sm}^{-1}$

$= 56.95 \times 10^{-2}\mathrm{Sm}^{-1} \approx 57 \times 10^{-2}\mathrm{Sm}^{-1}$

Q3 (45)

$$Fe^{3+} \xrightarrow{E_1^0} Fe^{2+} \xrightarrow{E_2^0} Fe$$

$$\underbrace{Fe^{3+} \longrightarrow Fe}_{E_3^0}$$

$$E_1^0 + 2E_2^0 = 3E_3^0$$

$$E_1^0 = 3E_3^0 - 2E_2^0$$

$= 3\ (-0.036) - 2(-0.44)$

$= +\ 0.772\ V$

$$E^0_{cell} = E^0_{Fe^{3+}/Fe^{2+}} + E^0_{I^-/I_2} = 0.233$$

$$\Delta_r G^0 = -\ 2 \times 96.5 \times 0.233 = -45\ kJ$$

Q4 (288)

From Kohlrausch's law

$$\Lambda_m^\infty (BaSO_4) = \lambda_m^\infty \left(Ba^{2+}\right) + \lambda_m^\infty \left(SO_4^{2-}\right)$$

$$\Lambda_m^\infty (BaSO_4) = \Lambda_m^\infty (BaCl_2) + \Lambda_m^\infty (H_2SO_4)$$

$$= 280 + 860 - 2(426)$$

$= 288 Scm^2\ mol^{-1}$